1 Team Problems

Problem 1. The points $(1, 2, 3)$ and $(3, 3, 2)$ are vertices of a cube. Compute the product of all possible distinct volumes of the cube.

Problem 2. Eight students attend a Harper Valley ARML practice. At the end of the practice, they decide to take selfies to celebrate the event. Each selfie will have either two or three students in the picture. Compute the minimum number of selfies so that each pair of the eight students appears in exactly one selfie.

Problem 3. Compute the least positive value of t such that

$$\operatorname{Arcsin}\big(\sin(t)\big),\ \operatorname{Arccos}\big(\cos(t)\big),\ \operatorname{Arctan}\big(\tan(t)\big)$$

form (in some order) a three-term arithmetic progression with a nonzero common difference.

Problem 4. In non-right triangle ABC, distinct points P, Q, R, and S lie on $\overline{BC}$ in that order such that $\angle BAP \cong \angle PAQ \cong \angle QAR \cong \angle RAS \cong \angle SAC$. Given that the angles of $\triangle ABC$ are congruent to the angles of $\triangle APQ$ in some order of correspondence, compute $\mathrm{m}\angle B$ in degrees.

Problem 5. Consider the system of equations

$$\begin{aligned}\log_4 x + \log_8(yz) &= 2\\ \log_4 y + \log_8(xz) &= 4\\ \log_4 z + \log_8(xy) &= 5.\end{aligned}$$

Given that xyz can be expressed in the form 2^k, compute k.

Problem 6. A complex number z is selected uniformly at random such that $|z| = 1$. Compute the probability that z and z^{2019} both lie in Quadrant II in the complex plane.

Problem 7. Compute the least positive integer n such that the sum of the digits of n is five times the sum of the digits of $(n + 2019)$.

Problem 8. Compute the greatest real number K for which the graphs of

$$(|x| - 5)^2 + (|y| - 5)^2 = K \quad \text{and} \quad (x - 1)^2 + (y + 1)^2 = 37$$

have exactly two intersection points.

Problem 9. To *morph* a sequence means to replace two terms a and b with $a+1$ and $b-1$ if and only if $a+1 < b-1$, and such an operation is referred to as a morph. Compute the least number of morphs needed to transform the sequence $1^2, 2^2, 3^2, \ldots, 10^2$ into an arithmetic progression.

Problem 10. Triangle ABC is inscribed in circle ω. The tangents to ω at B and C meet at point T. The tangent to ω at A intersects the perpendicular bisector of $\overline{AT}$ at point P. Given that $AB = 14$, $AC = 30$, and $BC = 40$, compute $[PBC]$.

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